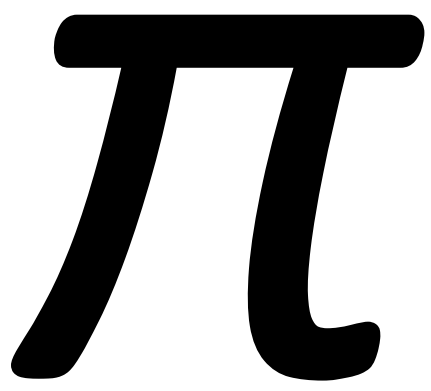
Estimating PI with Buffon's needle

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π



3.14159265358979323846264338327950288419716939937510582097494459230781640628620899862803482534211

- The first algorithms to calculate pi up to record-setting lengths of several million digits were based on formulae Srinivasa Ramanujan published in 1914.

- It helps figure out everything from calculating orbits to the size of paper rolls used in printers.



Credit: Joshua Fuller on Unsplash

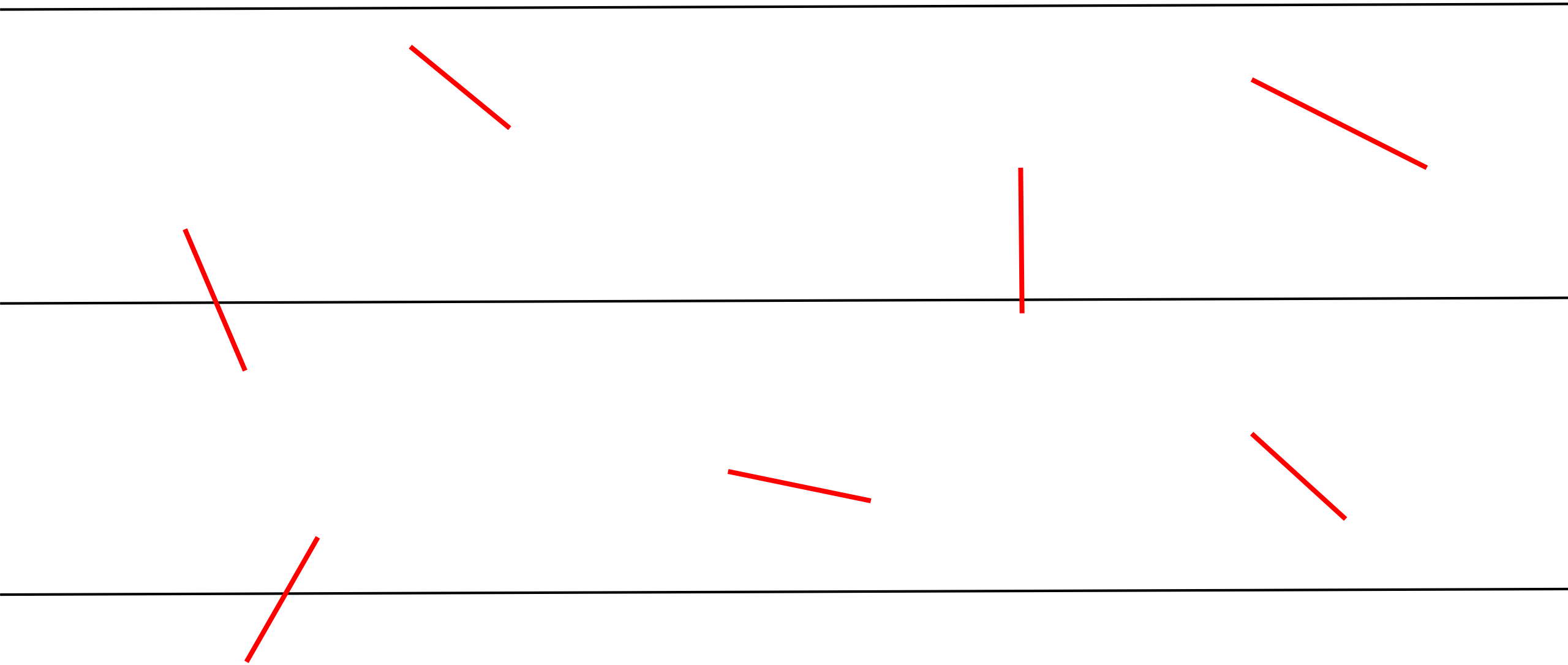
‘River Meandering as a Self-Organization Process’ - Hans-Henrik Stølum[1]
He proved that the average sinuosity of rivers around the world is pi. In other words, when you divide a rivers’ actual length by its’ straight route from its source to mouth, you get a number that is very close to pi.

[1]Department of Earth Sciences, University of Cambridge, Cambridge CB2 3EQ, UK.

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

It also appears in the Heisenberg's Uncertainty Principle equation.

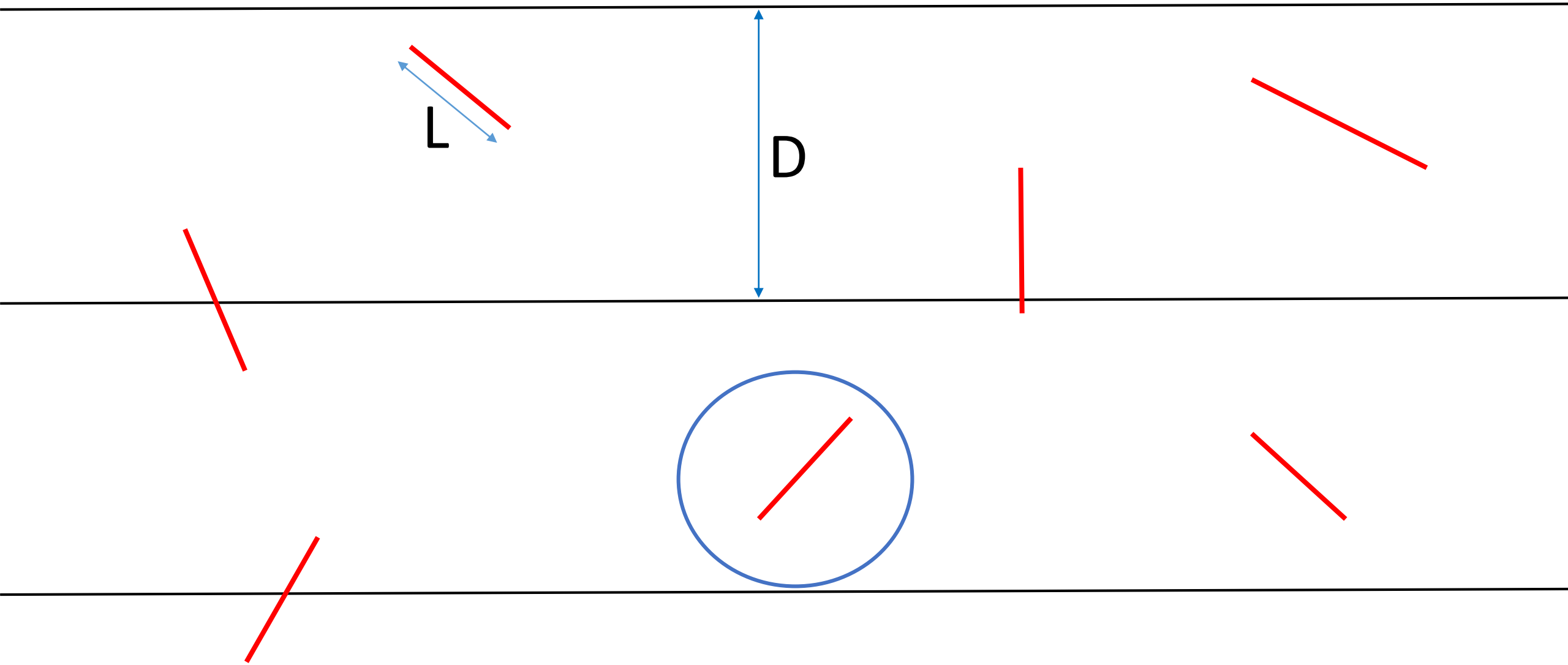
Buffon’s Needle Problem

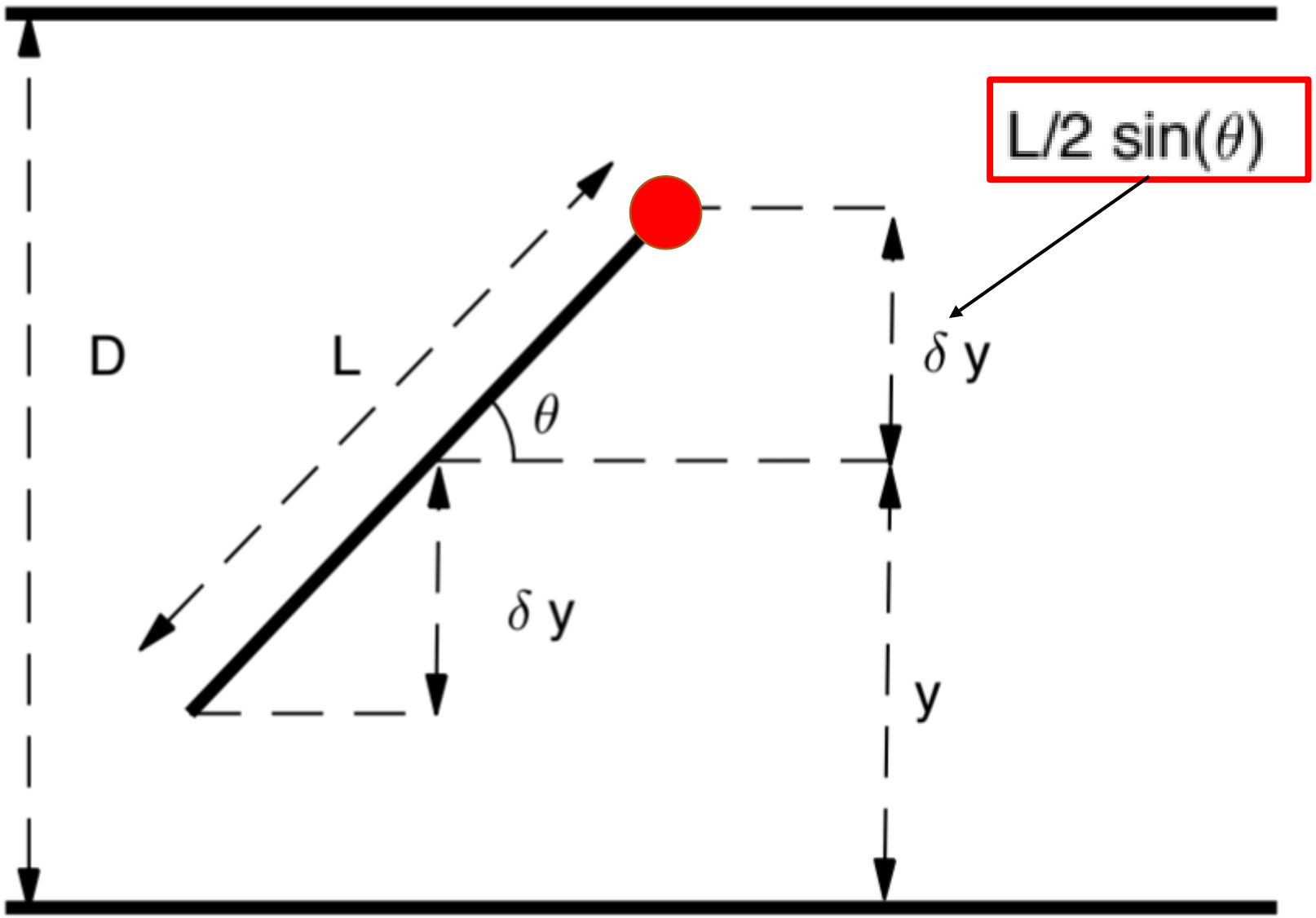


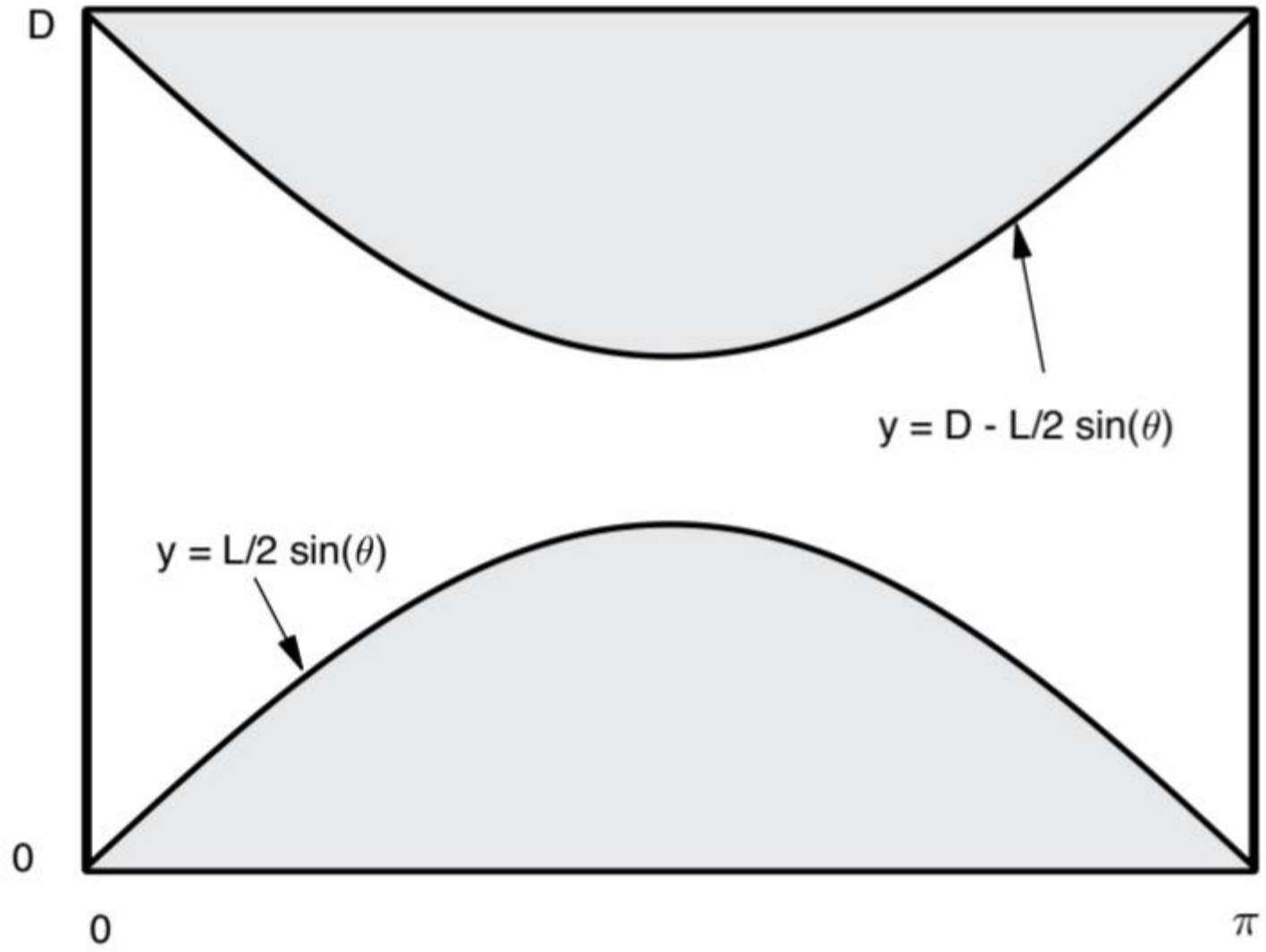


- One of the earliest known methods for empirically estimating the value of Pi (π).
- It is geometric probabilistic approach.
- It was first stated in 1777.
- It involves dropping a needle on a lined sheet of paper and determining the probability of the needle crossing one of the lines on the page.
- The remarkable result is that the probability is directly related to the value of pi.
- Today, we call this experiment ***"Buffon's Needle problem"*** to honor that French philosopher Georges-Louis Leclerc. And math people use integral geometry to explain this relation. You can find the proof below.

HITS?




L/2 sin(θ)
D
L
θ
δ y
δ y
y



Shaded portion = HITS

$$f = \frac{A}{\pi D}$$

$$A = 2\int_0^{\pi} \left(\frac{L}{2}\right) \sin\theta \, d\theta$$

$$f = \frac{2L}{\pi D}$$

$$\pi \approx \frac{2L}{fD}$$

Randomize θ and y

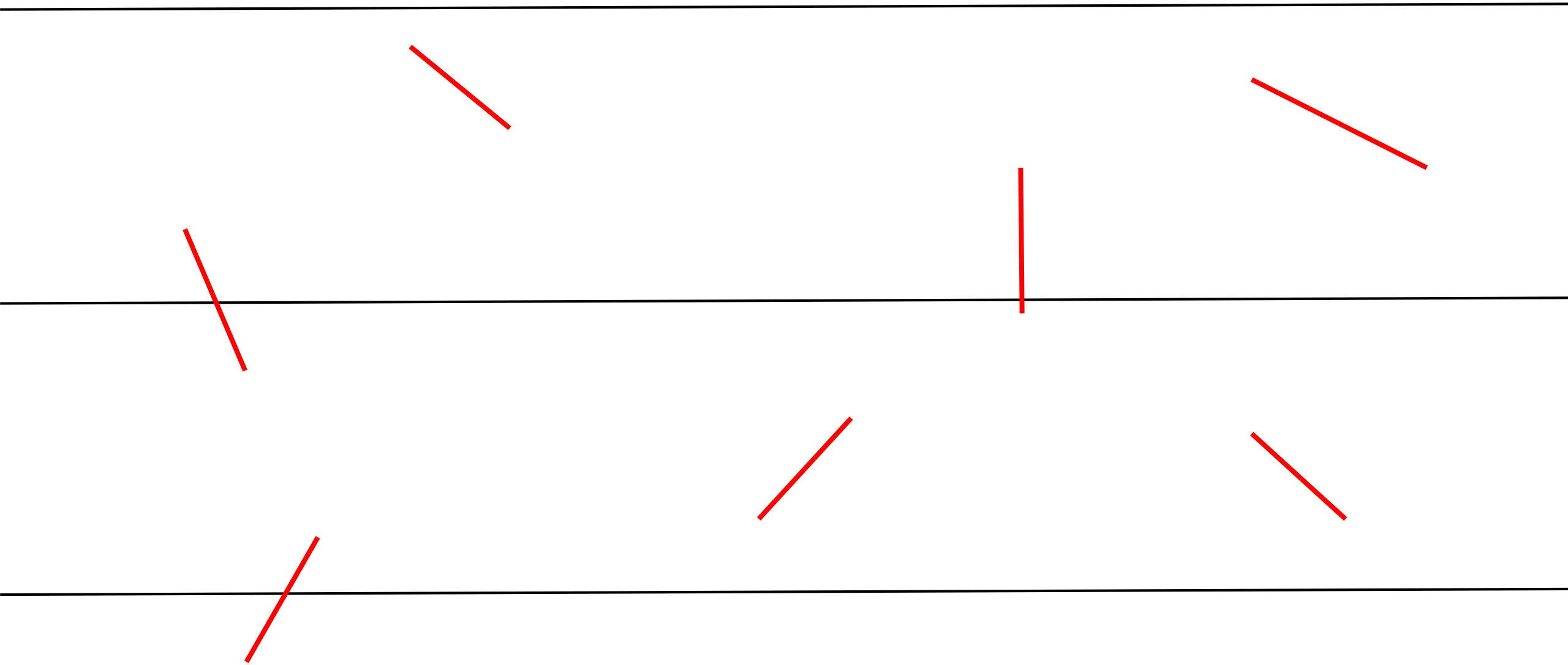
```
import random
import math

def buffon_needle_pi(num_trials, needle_length, line_spacing):
    hits = 0
    for _ in range(num_trials):
        angle_degrees = random.uniform(0, 90)
         angle_radians = math.radians(angle_degrees)
         distance = random.uniform(0, line_spacing / 2)
         y=(needle_length / 2) * math.sin(angle_radians)
         if distance <= y:
         hits += 1
    if hits == 0:
        return None
    return (2 * needle_length * num_trials) / (hits * line_spacing)

pi_estimate = buffon_needle_pi(1000000, 1.0, 1.0)
print("Buffon's Needle Pi Estimate:", pi_estimate)

print("Original value of Pi:", math.pi)
```

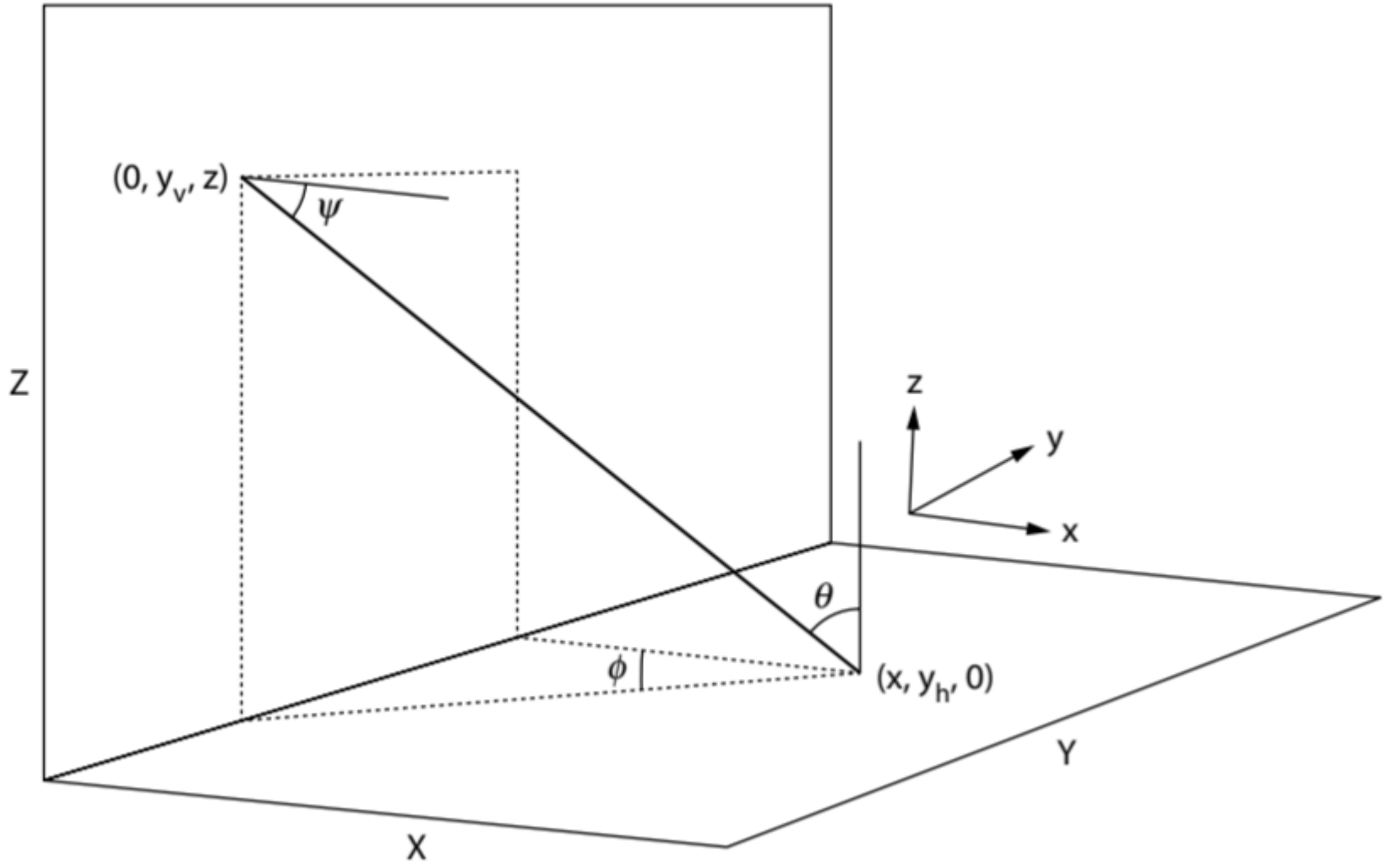
```
Buffon's Needle Pi Estimate: 3.142519546471579
Original value of Pi: 3.141592653589793
```



Monte Carlo Simulations

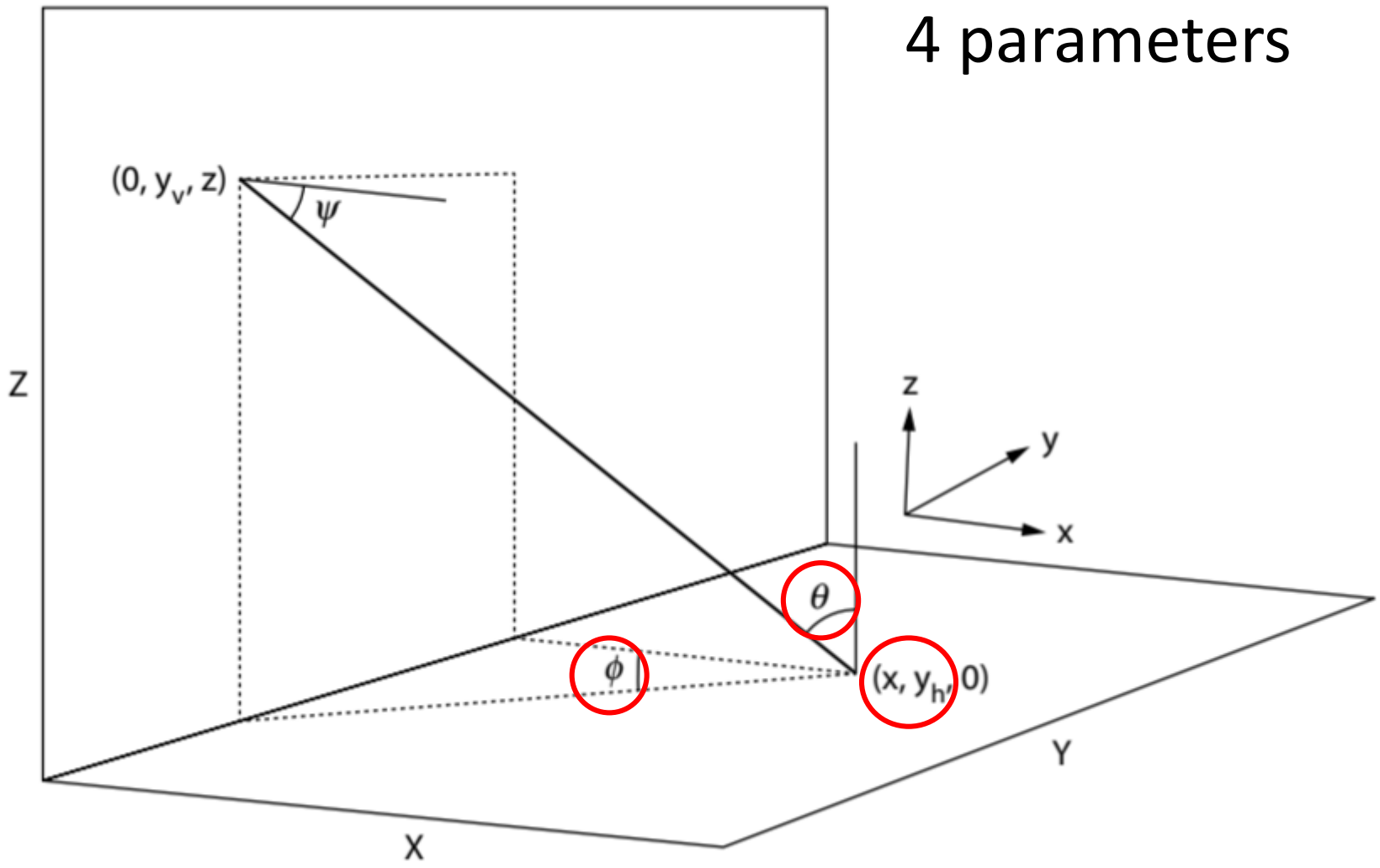
- Monte-Carlo simulation can be viewed as ‘experimental’ calculation, in which we use random numbers to conduct experiments.
- In reality, it is rarely a good idea to use a Monte-Carlo method to evaluate simple areas or one-dimensional integrals like those of our examples above.
- Conventional deterministic techniques are computationally much more efficient.
- In real scientific and engineering problems there could easily be several tens of inputs, giving rise to high dimensionality problems.
- Like the complex systems dynamics, where analytical solution might be too heavy where as by considering couple of parameters we can reach to a solution by randomly simulating them.

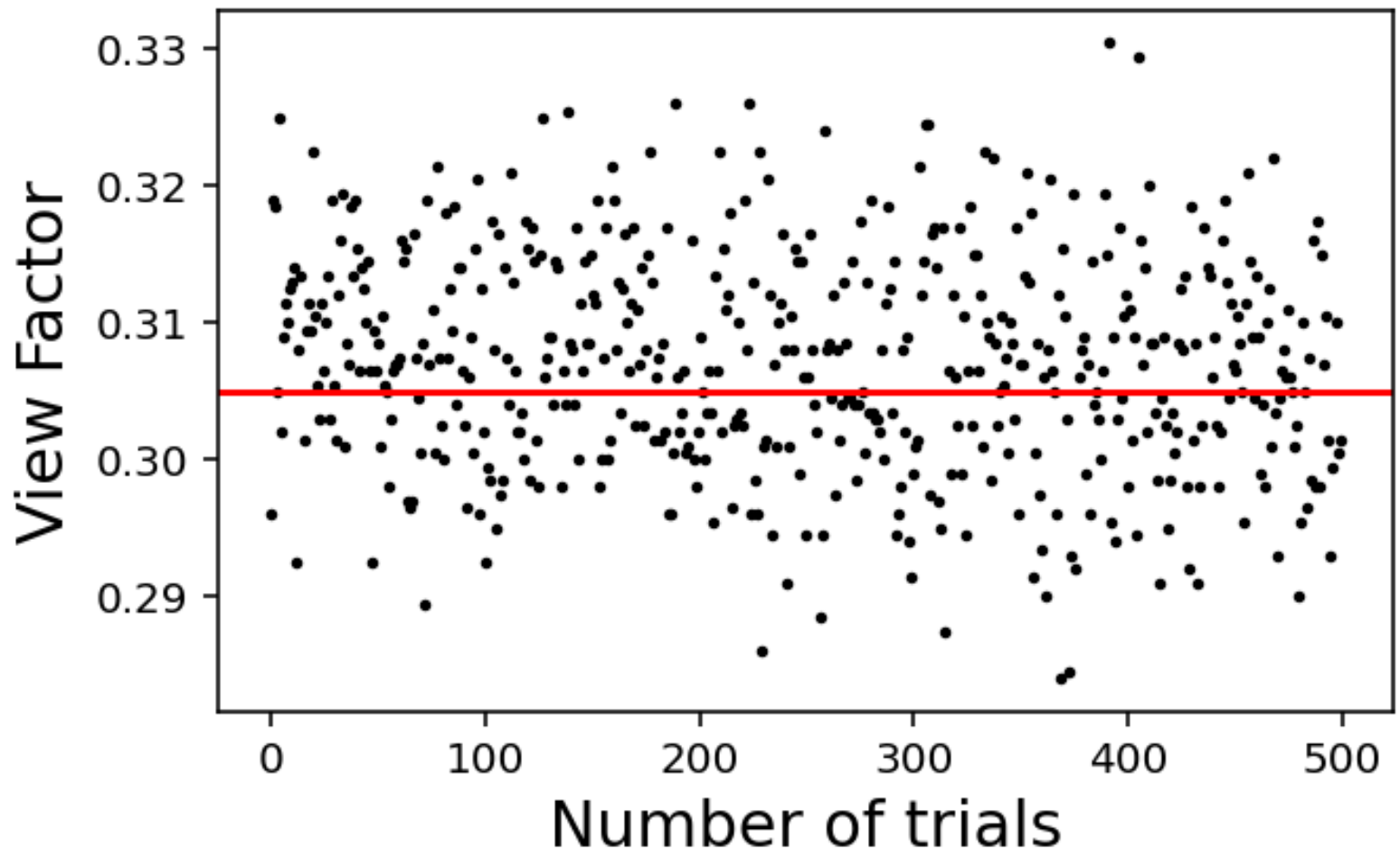
RADIATION VIEW FACTOR

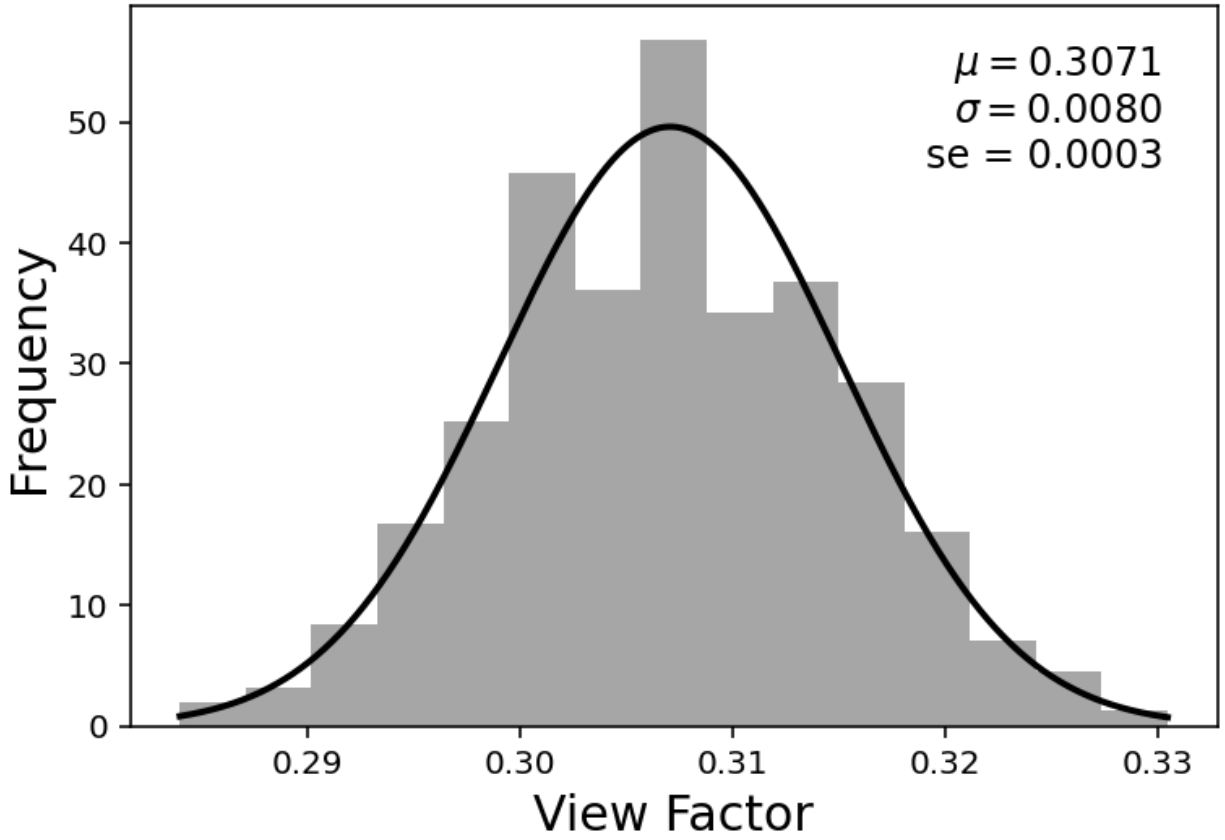


$$f_{12} = \frac{1}{A_1} \iint \frac{\cos\theta \cos\psi}{\pi r^2} dA_1 \, dA_2$$

4 parameters




0.33
0.32
0.31
0.30
0.29
View Factor
0
100
200
300
400
500
Number of trials



- We can now calculate the appropriate statistics and using the standard error on the mean to determine the lower and upper 95% limits.
- μ − 1.96se = 0.3065 and μ + 1.96se = 0.3075, respectively.
- These agree to two significant figures, so the best we can reasonably say (with 95% confidence) on the basis of the present calculation is that the view factor is about 0.30.

References

- Stevens, A. (2022). Monte-Carlo Simulation: An Introduction for Engineers and Scientists (1st ed.). CRC Press. https://doi.org/10.1201/9781003295235
- https://www.thehindu.com/sci-tech/science/pi-day-was-on-march-14-what-is-it-why-is-it-celebrated/article69338915.ece
- https://medium.com/however-mathematics/a-beautiful-way-to-calculate-%CF%80-buffons-needle-problem-51d41029f287
- https://arxiv.org/pdf/2402.06670
- https://mste.illinois.edu/activity/buffon/

Extra Readings

- Min et al., Buffon-Laplace Needle Problem as a geometric probabilistic approach to filtration process. December 2, 2024 https://arxiv.org/pdf/2402.06670.
- https://www.cmu.edu/mcs/news-events/2025/0314_pi-day.html
- Imitation Game